Code understanding journal -

1. Main components involved
Cli layer (cli.js)
- the purpose of this file is user-facing command interface using commander.js
- its responsibilities:
  * parse cli commands (create, list, status, delete, etc.)
  * format task output for terminal display
  * handle user input validation at cli level
  * display success/error messages
- key: does not contain business logic, only command routing

Business logic layer (app.js - taskmanager class)
- purpose: orchestrates all task operations
- responsibilities:
  * validate user inputs (e.g., date formats)
  * coordinate between cli and storage
  * implement business rules (e.g., special handling when marking as done)
  * calculate statistics and aggregate data
  * handle edge cases (invalid dates, missing tasks)
- key: acts as "manager" that delegates to storage

Data layer (storage.js - taskstorage class)
- purpose: manages persistence and in-memory storage
- responsibilities:
  * load tasks from tasks.json on startup
  * store tasks in memory as a hash (key = taskid, value = task object)
  * sync changes to disk (save to json)
  * provide query methods (filter by status, priority, overdue)
- key: dual responsibility - ram and disk persistence

Data model layer (models.js)
- purpose: defines data structures
- provides:
  * task class with properties and methods (markasdone(), isoverdue(), update())
  * taskpriority enum (1-4 levels)
  * taskstatus enum (todo, in_progress, review, done)
  * uuid generation for unique task ids
- key: contains business logic for individual tasks

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2. Execution flow - task creation
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Step-by-step flow:
1. User runs: node cli.js create "buy groceries" -d "from store x" -p 3

2. Cli parsing (cli.js)
  - commander.js intercepts the 'create' command
  - extracts arguments: title = "buy groceries"
  - extracts options: description, priority, due date, tags
  - parses tags string into array (if provided)

3. Taskmanager.createtask() (app.js)
  - validates date format if provided (catches yyyy-mm-dd format errors)
  - returns null if date invalid, console logs error
  - creates new task object: new task(title, description, priority, duedate, tags)

4. Task object creation (models.js)
  - constructor runs:
    * generates unique uuid (v4) for task.id
    * sets task.status = taskstatus.todo (default)
    * sets task.createdat = current timestamp
    * sets task.updatedat = createdat
    * all properties initialized (duedate, completedat=null, tags)

5. Storage.addtask() (storage.js)
  - stores task in memory: this.tasks[task.id] = task
  - returns the generated taskid

6. Storage.save() (storage.js)
  - converts this.tasks object to array
  - serializes to json (dates converted to iso strings automatically)
  - writes json to tasks.json file on disk
  - error handling: catches file i/o errors

7. Cli response (cli.js)
  - receives taskid from createtask()
  - prints: "created task with id: {taskid}"
  - user can now reference this task by id for future operations

Data state after creation:
- in ram: this.tasks = { "uuid-123": task{...}, ... }
- on disk: tasks.json contains json array with all tasks

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3. Execution flow - status update
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Step-by-step flow:
1. User runs: node cli.js status abc-def-ghi done

2. Cli parsing (cli.js)
  - extracts: taskid = "abc-def-ghi", status = "done"
  - calls: taskmanager.updatetaskstatus(taskid, status)

3. Taskmanager.updatetaskstatus() (app.js)
  - check: is newstatusvalue === taskstatus.done?

  if yes (marking as done):
    - retrieves task from storage: this.storage.gettask(taskid)
    - calls task.markasdone() (special handling!)
      * sets task.status = taskstatus.done
      * sets task.completedat = current timestamp (new!)
      * sets task.updatedat = completedat
    - calls storage.save() to persist changes
    - returns true

  if no (changing to other status):
    - uses standard updatetask path
    - calls storage.updatetask(taskid, { status: newstatusvalue })

4. Storage.updatetask() (storage.js - if not done)
  - gets task: task = this.tasks[taskid]
  - calls task.update(updates) - applies changes generically
  - calls storage.save()
  - returns true if successful, false if task not found

5. Task.update() (models.js - if not done)
  - iterates through updates object
  - sets each property on the task object
  - updates task.updatedat = new date()
  - (note: this is generic, doesn't know about status semantics)

6. Cli response (cli.js)
  - if updatetaskstatus returned true: prints "updated task status to {status}"
  - if false: prints "failed to update task status. Task not found."

Key difference from creation:

- creation: new object instantiated, added to collection
- update: existing object modified in place, then persisted
- special case: markasdone() records completion timestamp (for statistics)

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4. How data is stored and retrieved
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Storage architecture: in-memory hash + file-based persistence

On startup (taskstorage constructor):
1. Checks: does tasks.json file exist?
2. If yes:
  - read file: fs.readfilesync(storagepath, 'utf8')
  - parse json: json.parse(rawdata)
  - for each task in array:
    * create new task object (empty)
    * restore all properties: id, title, description, priority, status, tags
    * recreate date objects (json stored as strings, need to parse back)
      - createdat = new date(taskdata.createdat)
      - duedate = new date(taskdata.duedate) if exists
      - completedat = new date(taskdata.completedat) if exists
    * store in hash: this.tasks[task.id] = task
3. If no: this.tasks = {} (empty)

Why dates need special handling:
- json.stringify() converts date to iso string (e.g., "2026-04-24t10:30:00.000z")
- json.parse() does not convert string back to date object
- must explicitly call: new date(datestring) to recreate date objects
- without this: task.isoverdue() would fail (can't compare string < date)

During operations (read/write pattern):

Read (retrieve existing data):
- storage.gettask(taskid) → returns this.tasks[taskid]
- storage.getalltasks() → returns object.values(this.tasks)
- storage.gettasksbystatus(status) → filters this.tasks
- storage.gettasksbypriority(priority) → filters this.tasks
- storage.getoverduetasks() → filters with isoverdue() method

All reads are in-memory: extremely fast, no disk access

Write (modify and persist):
1. Modify in-memory object: this.tasks[id].property = value
2. Call storage.save()
   - convert tasks to array: object.values(this.tasks)
   - serialize to json: json.stringify(tasksarray, null, 2)
   - write to disk: fs.writefilesync(storagepath, jsonstring)
3. Error handling: catches and logs file i/o errors

Persistence guarantee:
- when save() completes successfully, data is on disk
- if program crashes, next startup loads from disk
- source of truth: tasks.json file (not ram)

Potential issue:
- if save() fails (disk full, permission error), changes only in ram
- next restart would lose those changes
- no transaction rollback mechanism

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5. Interesting design patterns discovered
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Pattern 1: layered architecture (3-tier design)
Structure: ui layer → business logic → data layer
Benefits:
  * separation of concerns (each layer has single responsibility)
  * cli can be replaced without touching business logic
  * easy to add rest api later (use same app.js, new api layer)
  * testing: can mock storage layer to test taskmanager

Pattern 2: in-memory cache with file sync
Pattern: load all data into ram, operate in ram, write back to disk
Benefits:
  * fast operations (all reads/writes in ram)
  * persistence (survives program restart)
  * simple (no database required)
Tradeoffs:
  * doesn't scale to millions of tasks
  * no concurrent access control (two instances could corrupt data)
  * single point of failure (tasks.json)

Pattern 3: hash-based lookup (this.tasks = {})
Pattern: store tasks in object with id as key instead of array

Benefits:
  * o(1) lookup by id: this.tasks[id] is instant
  * easier deletion: delete this.tasks[id]
  * no need to search array
Tradeoff:
  * must convert to array for saving/iteration: object.values(this.tasks)

Pattern 4: fluent command builder (commander.js)
Pattern: chain methods to build commands
Example: program.command('create').description(...).option(...).action(...)
Benefit:
  * declarative syntax is readable
  * consistent command definition across cli

Pattern 5: special-case handling in business logic
In updatetaskstatus(): special path for done status
Pattern: business rules that require extra side effects
Example: marking as done also sets completedat timestamp
Discovery: this is business logic, not generic property update
Pattern validates: task semantics belong in business layer (app.js), not data layer

Pattern 6: object reconstruction on load
In storage.load(): create new task objects from json data
Pattern: restore behavior and methods after deserialization
Discovery: task class methods like isoverdue() won't work on plain objects
Advantage: task behavior stays encapsulated in task class

Pattern 7: defensive date handling
Multiple places validate/convert dates:
  * app.js: validates user input format before creating task
  * storage.js: converts json strings back to date objects on load
Pattern: multiple validation layers catch errors at appropriate levels
Benefit: date errors handled gracefully, never crashes on comparison

Pattern 8: optional parameters with defaults
Task constructor: duedate = null, description = "", priority = 2, tags = []
Cli options: all have default values
Pattern: allows flexibility without requiring all fields
Benefit: users can create minimal tasks or detailed ones

Main insight:
This codebase demonstrates "crud with persistence" pattern:
- create: new task + save

- read: retrieve from in-memory hash
- update: modify object + save
- delete: remove from hash + save
Every mutation triggers save() - ensures disk is always consistent with ram state.

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6. Priority system - guided discovery learning
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Initial understanding (before guided questions):
- "priority is declared urgent when set to 4"
- "the '!' symbols are used to set priorities"
- "tasks are prioritized according to which has higher priority"
- confusion: mixing priority setting with priority display

Key misconception clarified:
The '!' symbols do not set priority - they display it!
- priority is set via the `-p` option during task creation (a number: 1, 2, 3, or 4)
- the '!' symbols are just visual formatting for terminal output (cosmetic)
- these are two completely separate concerns!

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Discovery #1: setting priority (input layer)
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Evidence from cli.js (line 52):
```
.option('-p, --priority <priority>', 'task priority (1-4)', 2)
```

The third parameter (2) is the default value.
- if user doesn't specify -p, priority defaults to 2 (medium)
- user specifies a number: -p 1, -p 2, -p 3, or -p 4

Important discovery: the value comes in as a string from cli
- user types: node cli.js create "task" -p 3
- arrives as: options.priority = "3" (string, not number)
- line 52 in cli.js shows passing to taskmanager.createtask(options.priority)

But in app.js line 10, it's used directly:
```
Const priority = priorityvalue;
Const task = new task(title, description, priority, duedate, tags);
```

```

Question exposed: is this a bug? Does priority get stored as string "3" or number 3?
Answer: look at models.js - the task constructor accepts it as-is
The system doesn't enforce type, so it works with both string and number
But this is loose typing - a potential source of bugs!

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Discovery #2: displaying priority (output/formatting layer)
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Evidence from cli.js lines 14-20:
```javascript
Const prioritysymbol = {
  [taskpriority.low]: '!',
  [taskpriority.medium]: '!!',
  [Taskpriority.high]: '!!!',
  [Taskpriority.urgent]: '!!!!'
};
```

The magic of [taskpriority.low]:
From models.js, we know:
```javascript
Const taskpriority = {
  low: 1,
  medium: 2,
  high: 3,
  urgent: 4
};
```

So [taskpriority.low] evaluates to:
```javascript
{
  [1]: '!',
  [2]: '!!',
  [3]: '!!!',
  [4]: '!!!!'
}
```

This creates an object that maps numbers to visual symbols:
```

- priority number 1 → displays as "!"
- priority number 4 → displays as "!!!!"

Critical insight: this object only formats display - it does not determine priority logic!
The actual priority ranking is determined by the number values (1 < 2 < 3 < 4)

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Discovery #3: how priority actually affects task ordering
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Initial misconception: "higher priority tasks appear first"
Reality check: let me trace where sorting happens...

Cli.js listtasks() with -p filter (line 65-72):
```javascript
If (priorityfilter) {
  return this.storage.gettasksbypriority(parseint(priorityfilter));
}
```

This filters to show only tasks of that priority level.
It does not sort by priority!

Storage.js gettasksbypriority() (line 60-61):
```javascript
Gettasksbypriority(priority) {
  return object.values(this.tasks).filter(task => task.priority === priority);
}
```

This returns tasks in iteration order (whatever order they're in the hash).

Discovery: there is no automatic sorting by priority anywhere in the codebase!

So if you run: node cli.js list
- tasks appear in order they were created, not by priority!
- the priority number is stored, but not used for ordering
- the '!' symbols show the priority visually, but don't drive any logic

Clarification: priority is stored but not used for ordering
The system supports priority as metadata, but doesn't implement priority-based sorting.

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Discovery #4: full priority data flow
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Complete journey of priority value:

1. User input
  node cli.js create "task" -p 3

2. Cli parsing (cli.js)
  options.priority = "3" (string from commander.js)

3. Business logic (app.js)
  taskmanager.createtask(title, desc, "3", ...)
  const priority = priorityvalue; // priority = "3"
  new task(..., priority, ...)

4. Data model (models.js)
  task.priority = "3" (stored as-is, type not enforced!)

5. Storage (storage.js)
  this.tasks[taskid].priority = "3"
  saved to json as: { priority: "3" }

6. Display (cli.js formattask)
  parseint("3") doesn't happen explicitly... But prioritysymbol["3"] might fail!
  actually, this is another bug - formattask assumes priority is number but it's stored as string

  when displayed:
  prioritysymbol["3"] = undefined (not "3")
  falls back to nothing... Task would display without !!! Symbols!

Actual bug discovered: priority type inconsistency causes display issues!

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Discovery #5: what actually works correctly
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Despite the type inconsistency, here's what does work:

Filtering by priority works because:
- storage.gettasksbypriority(priority) does: task.priority === priority

- since both sides are strings (or both numbers), === comparison works

Display currently might have issues, but users probably:
- create tasks with number for -p (node cli.js create "x" -p 3)
- this passes as string "3"
- internally stays as "3"
- but formattask uses prioritysymbol[task.priority]
- prioritysymbol[3] works if priority somehow became number 3

Question: how does it work then if there's a type mismatch?
Answer: the code is fragile but hasn't broken because:
- priority filtering works with string comparison
- display might be using implicit type coercion
- in practice, javascript's loose typing masks the issue

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Misconceptions clarified
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Misconception #1: "! Symbols set the priority"
Correction: ! Symbols display the priority (formatting only)
Priority is set by the -p option (a number)

Misconception #2: "4 means urgent so tasks with 4 display first"
Correction: priority is stored but not used for sorting
The system doesn't implement priority-based ordering
Tasks display in creation order regardless of priority level

Misconception #3: "priority affects task querying/filtering"
Correction: priority affects filtering (showing only one level)
But doesn't affect sorting (order within results)
A list query with no filter shows all tasks in creation order

Misconception #4: "priority system is complete and correct"
Correction: the system has type inconsistency issues
Priority comes in as string but should be number
Formattask() assumes numeric priority but might receive string
This works in practice due to javascript's loose typing, but is fragile

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Mental model correction
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Before: priority = "a ranking system that determines task order"
After: priority = "metadata field that categorizes tasks by urgency level"

The system stores priority but doesn't actively use it for ordering.
It's like filing tasks in folders labeled "urgent", "high", etc.
But if you ask for "all tasks", you get them in filing order, not by urgency.

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Practical implications
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If i wanted to modify the code to show highest priority tasks first:

Option 1 (quick fix): modify storage.getalltasks() to sort
```
Return object.values(this.tasks).sort((a, b) => b.priority - a.priority);
```
(sorts descending: 4, 3, 2, 1)

Option 2 (better fix): add separate method in storage for sorted retrieval
```
Gettaskssortedbypriority() { ... }
```

Option 3 (best fix): fix type consistency first
- ensure priority is always stored as number
- update line 52 in cli.js to: parseint(options.priority)
- then add sorting logic

Key insight: the code works despite loose typing, but adding sorting would expose the type inconsistency bug.

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7. Data flow diagram - task completion
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Visual data flow:

```
User terminal
    |
    | "node cli.js status abc-def-ghi done"
    v
 ┌──────────────────────────────────────────────────────┐
```

```
│  cli layer (cli.js)  │  - parse command
└──────────────────────┴──────────────────────┘  - extract taskid, status
        |
        v
┌──────────────────────────────────────────────────────────────────────┐
│ business logic (app.js)          │  - check if done status
│ updatetaskstatus()               │  - route to special handler
└──────────────────────┴──────────────────────────────────────────────┘
        |
        v
┌──────────────────────────────────────────────────────────────────────┐
│ data layer (storage.js)          │  - retrieve task from ram
│ storage.gettask(taskid)          │  - in-memory hash lookup
└──────────────────────┴──────────────────────────────────────────────┘
        |
        v
┌──────────────────────────────────────────────────────────────────────┐
│ task model (models.js)           │  - call markasdone()
│ task.markasdone()                │  - set status = "done"
└──────────────────────┴──────────────────────────────────────────────┘
- set completedat = now
        |
        v
┌──────────────────────────────────────────────────────────────────────┐
│ storage layer (storage.js)       │  - convert task → json
│ storage.save()                   │  - write to tasks.json
└──────────────────────┴──────────────────────────────────────────────┘
        |
        v
┌──────────────────────────────────────────────────────────────────────┐
│  file system (tasks.json)        │  - persist to disk
│  synchronous write               │  - data survives restart
└──────────────────────────────────────────────────────────────────────┘
```

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8. State changes during task completion
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State transition:

Before completion:
```javascript
{
  id: "abc-def-ghi",
  title: "buy groceries",
  status: "todo",
  completedat: null,
  updatedat: "2026-04-20t10:30:00.000z",  // when created
  priority: 3,
  duedate: "2026-04-25t00:00:00.000z",
  tags: ["shopping"],
  // ... Other properties
}
```

After markasdone() called:
```javascript
{
  id: "abc-def-ghi",
  title: "buy groceries",
  status: "done",                    // changed
  completedat: 2026-04-24t14:35:22.123z, // new - current time
  updatedat: 2026-04-24t14:35:22.123z,   // changed - timestamp updated
  priority: 3,
  duedate: "2026-04-25t00:00:00.000z",
  tags: ["shopping"],
  // ... Other properties unchanged
}
```

Locations of state:

Location 1 - ram (task object in memory):
- before: task.status = "todo", task.completedat = null
- after: task.status = "done", task.completedat = date object
- speed: instant

- visibility: only in this process
- persistence: lost if program crashes before save()

Location 2 - disk (tasks.json):
- before: "status": "todo", "completedat": null
- after: "status": "done", "completedat": "2026-04-24t14:35:22.123z"
- speed: depends on disk i/o (milliseconds)
- visibility: can view file in text editor
- persistence: survives any crashes

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9. Potential points of failure
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Failure point 1: task not found
- location: app.js updatetaskstatus()
- detection: storage.gettask(taskid) returns undefined
- handling: returns false, cli shows error message
- user experience: ✓ good - clear error
- example: user types wrong task id

Failure point 2: task id formatting issue
- location: cli.js command parsing
- detection: taskid contains special characters that don't match
- handling: none - accepts any string
- user experience: ✗ bad - silently fails
- example: copy-paste error in task id

Failure point 3: status value invalid
- location: app.js updatetaskstatus()
- detection: newstatusvalue is not a valid taskstatus value
- handling: none - accepts any string like "done2" or "done"
- user experience: ✗ bad - creates corrupt data
- example: typo: node cli.js status abc done1

Failure point 4: date creation fails (rare)
- location: models.js markasdone() → new date()
- detection: javascript date() returns invalid date
- handling: none - just stores invalid date object
- user experience: ✗ bad - crashes later on date comparison
- example: system clock corrupted

Failure point 5: disk write fails (silent!)
- location: storage.js save() → fs.writefilesync()
- detection: fs.writefilesync() throws error (no disk space, permissions, etc.)
- handling: caught in try-catch, logged to console only
- user experience: ✗ critical - user sees "updated!", but data lost!
- example: disk full, file locked by antivirus, read-only filesystem
- consequence: next restart loses changes, user thinks task was saved

Failure point 6: file corrupted during read on restart
- location: storage.js load() → json.parse()
- detection: tasks.json is corrupted (malformed json)
- handling: caught in try-catch, app starts with empty task list
- user experience: ✗ bad - silent data loss
- example: disk error mid-write, manual file edit with syntax error
- consequence: all tasks disappear without warning

Failure point 7: task reference gone stale
- location: between multiple commands
- detection: task modified by another process/instance
- handling: none - single-process app, but no concurrency control
- user experience: ✗ bad - if two instances run, data gets corrupted
- example: running two cli instances simultaneously

Failure point 8: completedat timestamp wrong
- location: models.js markasdone() → new date()
- detection: system clock changes during execution
- handling: none - uses whatever time is returned
- user experience: ~ medium - timestamp might be incorrect but not critical
- example: ntp sync happens, time jumps backward

Critical gaps in error handling:
- ✕ no validation that save() succeeded
- ✕ no user feedback if write fails
- ✕ no validation of status enum values
- ✕ no validation of completedat timestamp
- ✕ no file locking (multi-instance corruption risk)
- ✕ no backup before save
- ✕ no rollback on failure
- ✕ no retry logic for transient failures

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10. Persistence mechanism
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How changes are persisted:

Step 1: trigger point
```javascript
// in app.js updatetaskstatus():
Task.markasdone();       // modifies task object in ram
This.storage.save();     // triggers persistence
```

Step 2: preparation (in memory)
```javascript
// in storage.js save():
Const tasksarray = object.values(this.tasks);
// now we have: [ { id, title, status, completedat, ... }, ... ]
```

Step 3: serialization (ram → string)
```javascript
Json.stringify(tasksarray, null, 2)
// date objects automatically converted to iso strings:
// "2026-04-24t14:35:22.123z"
// special formatting (null, 2) makes file human-readable
```

Step 4: write to disk (string → file)
```javascript
Fs.writefilesync(this.storagepath, jsonstring)
// synchronous call - blocks until written
// returns when bytes are flushed to storage device
```

Step 5: verification (optional but missing)
```javascript
// missing in current code:
// if (save succeeded) { return true } else { alert user }
```

Persistence guarantees:

✓ guaranteed if:
- save() completes without throwing error
- file system is functioning normally
- no power loss during writefilesync()

✗ not guaranteed if:
- save() throws error (caught but ignored)
- disk full during write (partial write, corrupted file)
- file permissions change
- another process locks the file
- program receives sigkill signal
- disk i/o timeout occurs

Data recovery after failure:

Scenario 1: program crashes before save()
- result: changes lost, file on disk unchanged
- next restart: old data loads (good)

Scenario 2: program crashes during save()
- result: tasks.json partially written (corrupted!)
- next restart: json.parse() fails, app starts empty (bad!)
- recovery: manual restore from backup (no backup system)

Scenario 3: program crashes after save()
- result: changes persisted
- next restart: reloads completed tasks (good)

Flow summary:

Ram state change (instant)
↓
Save() called
↓
Json serialization
↓
Disk write (blocking i/o)
↓
File on disk updated

↓

Next restart loads from disk

No intermediate steps between ram and disk
No atomic transactions (can't roll back)
No backup before write (data can be lost)